Monte Carlo Simulation for CT-Based Radiation Dose Planning Using Patient-Specific Tissue Modeling

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***Abstract*—Accurate radiation dose estimation is fundamental to effective radiotherapy treatment planning. This study presents a computational framework for simulating radiation dose deposition using patient-specific computed tomography (CT) images. The approach integrates deterministic beam modeling based on exponential attenuation and Monte Carlo photon transport simulation to capture both macroscopic attenuation and stochastic scattering effects.**

CT images are converted into Hounsfield Units (HU), segmented into tissue classes, and mapped to material-specific attenuation coefficients. A comparative analysis shows that Monte Carlo simulation produces more realistic and heterogeneous dose distributions, particularly in the presence of tissue heterogeneity.

Targeted beam delivery increased mean tumor dose from 0.088 to 3.59 (non-normalized units), although spatial dose variability remained significant, with an underdose probability of 0.398.

I. Introduction

Radiation therapy is a cornerstone in cancer treatment, where accurate dose delivery is critical for maximizing tumor control while minimizing damage to healthy tissues. Dose calculation plays a central role in treatment planning, as inaccuracies can lead to underdosing of tumors or unnecessary exposure of surrounding organs.

In this work, dose values are expressed in arbitrary units proportional to relative energy deposition and are used for comparative analysis rather than absolute clinical calibration.

Conventional dose calculation approaches often rely on simplified physical models, such as analytical attenuation or convolution-based methods. While computationally efficient, these approaches struggle to accurately represent photon interactions in heterogeneous anatomical structures, particularly in regions with sharp density transitions such as bone–soft tissue interfaces.

Monte Carlo simulation is widely regarded as the gold standard for radiation transport modeling due to its ability to simulate stochastic photon interactions, including scattering and probabilistic energy deposition. However, its high computational cost limits its routine use in many practical settings.

In this work, a computational framework was developed that integrates CT-based tissue modeling with both deterministic and Monte Carlo radiation transport methods. Patient-specific CT data is processed into Hounsfield Units (HU), segmented into material classes, and mapped to attenuation coefficients to enable physics-based simulation of dose deposition.

The objective of this study is to:

- Simulate radiation dose deposition using real CT image data
- Compare deterministic attenuation-based modeling with Monte Carlo simulation
- Analyze the impact of tissue heterogeneity on dose distribution
- Evaluate the effectiveness of targeted beam delivery for tumor irradiation

This work emphasizes interpretability and physical consistency, providing a transparent framework for understanding how imaging-derived tissue properties influence radiation dose deposition.

II. Related Work

Recent advancements in radiotherapy simulation have focused on improving accuracy through Monte Carlo methods and patient-specific modeling. Studies such as Kawrakow et al. (2018) demonstrated improved dose accuracy using GPU-based Monte Carlo simulations.

Other works have explored deep learning approaches for dose prediction; however, these models often lack interpretability and physical consistency.

This work combines physically grounded simulation with real imaging data, providing a transparent and interpretable framework.

III. System Overview

The proposed framework is organized as a modular computational pipeline that transforms CT image data into simulated radiation dose distributions. Each stage of the pipeline corresponds to a distinct processing or modeling task, enabling clear separation between imaging, physics, and analysis components.

The system consists of five core modules:

- **Data Acquisition**: Loading CT images in DICOM format

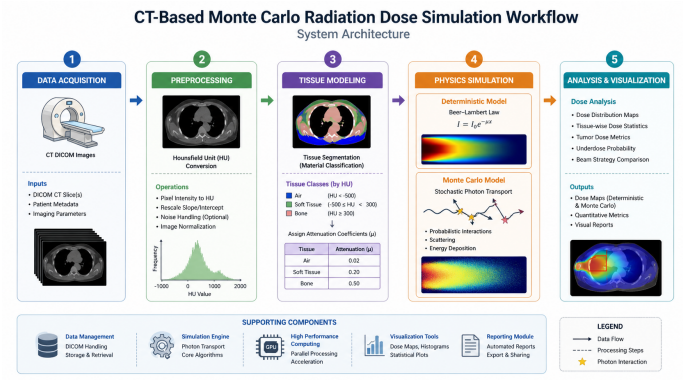

Fig. 1. High-level system architecture of CT-based Monte Carlo radiation dose simulation.

- **Preprocessing**: Conversion of pixel intensities to Hounsfield Units (HU)
- **Tissue Modeling**: Segmentation into material classes and assignment of attenuation coefficients
- **Physics Simulation**: Dose computation using deterministic attenuation and Monte Carlo photon transport
- **Analysis Layer**: Statistical evaluation and visualization of dose distributions

This modular design allows individual components to be modified or extended independently. In particular, the framework can be expanded to support 3D volumetric simulations, more detailed tissue classification, or advanced radiation transport models.

NOMENCLATURE

Symbol	Description
HU	Hounsfield Unit
μ	Linear attenuation coefficient
I_0	Initial beam intensity
x	Path length
D	Dose
$\mathcal{T}$	Tumor region
σ	Standard deviation
ξ	Random variable $\in (0, 1)$

IV. METHODOLOGY

A. CT Image Processing

CT images were obtained in DICOM format and converted into Hounsfield Units (HU) using scanner-specific calibration parameters:

$$HU = p \cdot \text{slope} + \text{intercept} \tag{1}$$

where p represents the raw pixel intensity, and *slope* and *intercept* are rescaling parameters extracted from the DICOM metadata (RescaleSlope and RescaleIntercept).

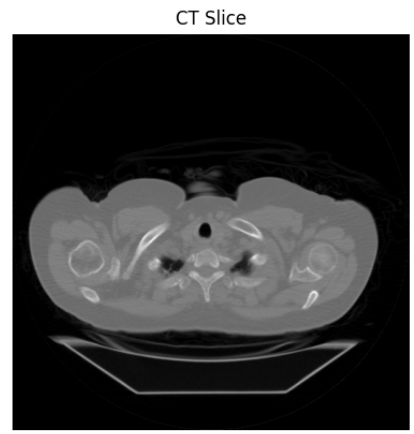

Fig. 2. CT Image Slice

The CT image provides spatial density variations that directly influence radiation attenuation and interaction probability.

Hounsfield Units provide a normalized measure of radiodensity, where air is approximately -1000 HU, water is 0 HU, and dense bone can exceed $+1000$ HU. These values directly influence photon attenuation and interaction probability in the simulation.

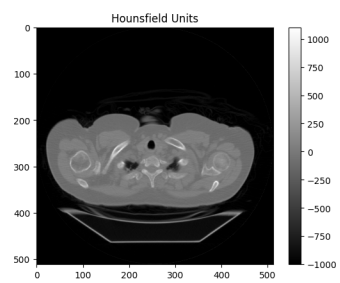

Fig. 3. Hounsfield Unit reference visualization

B. Tissue Segmentation

Tissue classification was performed using threshold-based segmentation on Hounsfield Unit (HU) values:

- Air: $HU < -500$
- Soft Tissue: $-500 \leq HU < 300$
- Bone: $HU \geq 300$

These thresholds provide a simplified approximation of anatomical structures and are commonly used for coarse material separation. Each tissue class was assigned a corresponding linear attenuation coefficient (μ):

The attenuation coefficient μ determines the probability of photon interaction per unit path length and varies with both material composition and photon energy. In this work, fixed representative values were used to simplify the simulation while preserving relative differences between tissue types.

Tissue	Attenuation (μ)
Air	0.02
Soft Tissue	0.20
Bone	0.50

TABLE I
ASSIGNED ATTENUATION COEFFICIENTS FOR TISSUE CLASSES

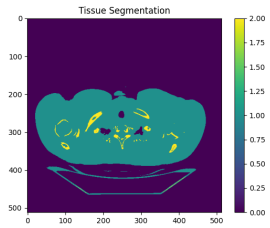

Fig. 4. Tissue Segmentation Map

A simplified mapping was used, assuming a monotonic relationship between Hounsfield Units and linear attenuation coefficients.

C. Deterministic Beam Model

The deterministic model estimates photon attenuation using the Beer–Lambert law:

$$I(x) = I_0 e^{-\mu x} \tag{2}$$

where I_0 is the initial beam intensity, μ is the linear attenuation coefficient of the traversed tissue, and x represents the path length through the material.

Energy deposition is approximated as the fraction of attenuated beam intensity:

$$D(x) = I_0 \left(1 - e^{-\mu x}\right) \tag{3}$$

This formulation assumes that absorbed dose is proportional to the fraction of attenuated photon energy, neglecting secondary particle transport.

In this model, photons are assumed to travel in straight paths without scattering. As a result, dose decreases smoothly with depth and does not exhibit lateral spread.

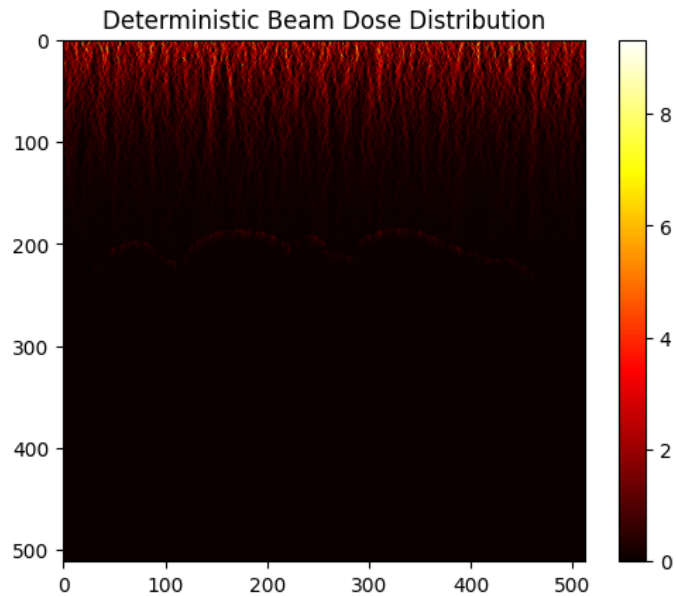

Fig. 5. Deterministic Dose Distribution

The smooth gradient confirms Beer–Lambert attenuation behavior.

- High dose near entry point
- Rapid attenuation in bone
- Limited lateral spread

D. Monte Carlo Photon Transport

Photon transport is modeled using stochastic sampling of interaction events. At each step, the probability of interaction is determined by:

$$P_{interact} = 1 - e^{-\mu \Delta x} \tag{4}$$

where μ is the attenuation coefficient of the local tissue and Δx is the step size.

Alternatively, photon step lengths can be sampled using an exponential distribution:

$$s = -\frac{\ln \xi}{\mu} \tag{5}$$

where $\xi \in (0, 1)$ is a uniformly distributed random number. This formulation models the mean free path between successive interactions.

At each interaction, energy is deposited locally, and the photon may undergo scattering, resulting in a change of direction. This stochastic process produces heterogeneous dose distributions and lateral spread that are not captured by deterministic models. Photon energy was assumed to be mono-energetic, and energy-dependent interaction cross-sections were not explicitly modeled.

Algorithm 1 Monte Carlo Photon Transport

```
for each photon do
    Initialize position and energy
    while energy > threshold do
        Sample step length s
        Move photon forward by s
        Determine interaction
        if interaction occurs then
            Deposit energy locally
            Reduce photon energy
            Update photon direction (scattering)
        end if
    end while
end for
```

This algorithm represents a simplified photon transport model and does not explicitly simulate interaction types such as Compton scattering or photoelectric absorption.

V. Implementation Details

The simulation framework was implemented in Python using NumPy for numerical computation and Matplotlib for visualization. CT image data was processed using the pydicom library to extract pixel intensities and associated metadata (e.g., RescaleSlope and RescaleIntercept).

The computational pipeline operates on a 2D CT slice represented as a grid, where each pixel corresponds to a tissue-dependent attenuation coefficient. Deterministic dose computation was performed using vectorized operations over the grid, enabling efficient evaluation of exponential attenuation.

The Monte Carlo simulation was implemented using iterative photon tracking, where each photon undergoes stochastic propagation, interaction, and energy deposition. Random number generation was used to sample step lengths and interaction events, introducing statistical variability in the resulting dose distribution.

A. Computational Complexity

Let M denote the number of pixels in the CT image and N the number of simulated photons.

- Deterministic model complexity: $O(M)$, as attenuation is computed once per pixel.
- Monte Carlo model complexity: $O(N \cdot H)$, where H represents the average number of interaction steps per photon.

The deterministic model is computationally efficient due to its direct evaluation over the image grid. In contrast, the Monte Carlo simulation requires repeated sampling and iterative photon tracking, resulting in significantly higher computational cost.

In practice, simulation time increases linearly with the number of photons, making accuracy–performance trade-offs necessary when selecting N.

The Monte Carlo simulation is highly parallelizable, as photon histories are independent, making it suitable for GPU acceleration.

B. Simulation Trade-off

The two models represent a trade-off between:

- **Speed**: Deterministic model is fast but simplified
- **Accuracy**: Monte Carlo model is realistic but computationally intensive

VI. Experimental Setup

Simulations were performed on a 2D chest CT slice, where each pixel corresponds to a tissue-dependent attenuation coefficient derived from Hounsfield Units.

A photon beam was initialized at the image boundary and directed toward the interior of the domain. For stochastic simulations, photon starting positions were sampled from a Gaussian distribution centered along the beam axis, with standard deviation $\sigma = 3$, controlling beam spread.

A tumor region was defined within a soft tissue area to evaluate targeted dose delivery. The region was selected to lie away from high-density bone structures to isolate tissue interaction effects.

The main simulation parameters were:

- Number of photons: 15,000
- Beam spread (standard deviation): $\sigma = 3$
- Grid resolution: equal to CT image resolution

VII. Results

A. Dose Distribution

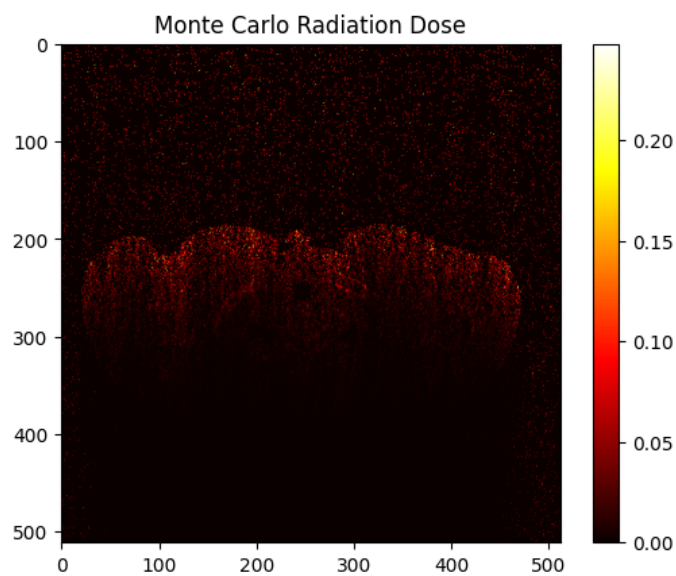

Fig. 6. Monte Carlo dose distribution across the CT slice

Figure 6 shows the spatial distribution of radiation dose obtained using the Monte Carlo simulation. The dose is highest near the beam entry region and decreases with depth due to attenuation.

Lateral spread of dose is observed across neighboring regions, indicating photon scattering during transport. Compared to the deterministic model (Figure 5), the Monte Carlo simulation produces a broader and more heterogeneous distribution.

Soft tissue regions exhibit higher dose accumulation, while bone structures attenuate photon propagation, resulting in reduced dose beyond dense regions. This behavior reflects the dependence of photon interaction probability on tissue composition.

B. Tissue Dose Analysis

Tissue	Deterministic	Monte Carlo
Air	0.275	0.029
Soft Tissue	0.010	0.131
Bone	≈ 0	0.006

TABLE II
DOSE COMPARISON

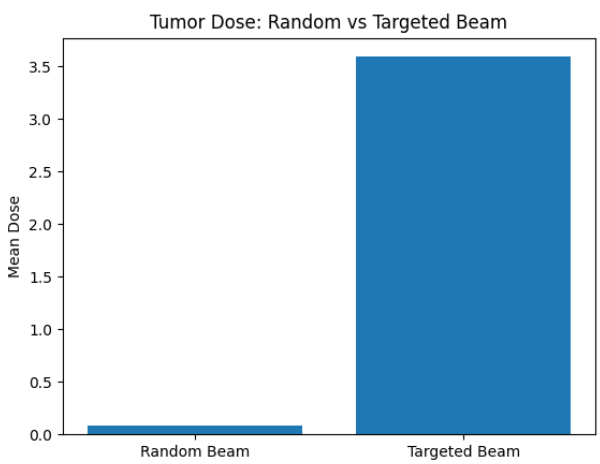

Fig. 7. Targeted vs Random Beam

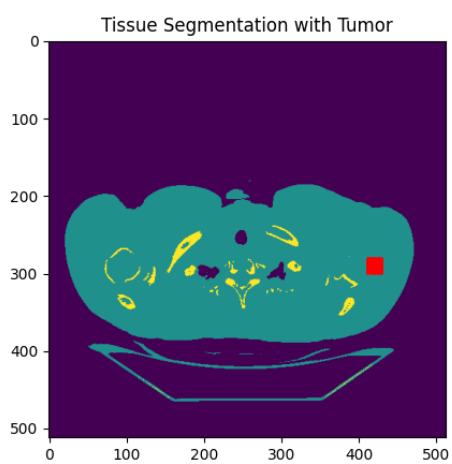

Fig. 8. Tumor Location

1) Tumor Region Definition: The tumor region $\mathcal{T}$ was defined within soft tissue as a square region centered at (i_c, j_c):

$$\mathcal{T} = \{(i,j) \mid |i - i_c| \leq r,\ |j - j_c| \leq r\} \tag{6}$$

where $r = 10$ pixels. The results indicate that while targeted beam delivery significantly increases the mean tumor dose, the spatial distribution remains heterogeneous. Regions within the tumor exhibit varying dose intensities, which may lead to partial treatment inefficiency. This highlights the importance of optimizing beam configurations and incorporating multiple beam angles in clinical applications to achieve uniform dose coverage.

2) Dose Normalization: To allow comparison across simulations, the dose map was normalized:

$$D_{norm}(i,j) = \frac{D(i,j)}{\max(D)} \tag{7}$$

3) Tumor Dose Statistics: Statistical analysis of the normalized tumor dose yielded:

- Mean dose: $\bar{D} = 0.33$
- Standard deviation: $\sigma = 0.086$
- Maximum dose: $D_{max} = 0.57$
- Minimum dose: $D_{min} = 0.16$

This corresponds to an increase of over 40× in mean tumor dose, highlighting the effectiveness of spatial beam targeting.

4) Underdose Probability: The probability of underdose within the tumor region was computed as:

$$P_{under} = \frac{1}{N} \sum_{(i,j)\in\mathcal{T}} I(D_{norm}(i,j) < D_{th}) \tag{8}$$

Using a threshold $D_{th} = 0.3$, the underdose probability was:

$$P_{under} = 0.398 \tag{9}$$

5) Dose Distribution: The distribution of normalized dose values within the tumor is shown in Figure 9. The histogram illustrates the spread of dose values across the tumor region.

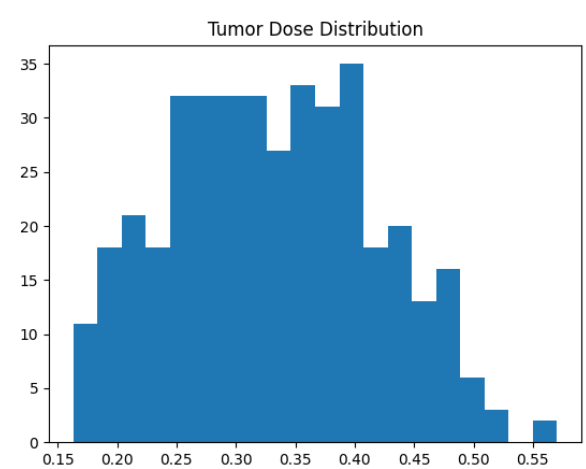

Fig. 9. Normalized tumor dose distribution

6) Spatial Dose Distribution: The spatial distribution of dose across the tumor region is shown in Figure 10. The tumor boundary is overlaid on the Monte Carlo dose map.

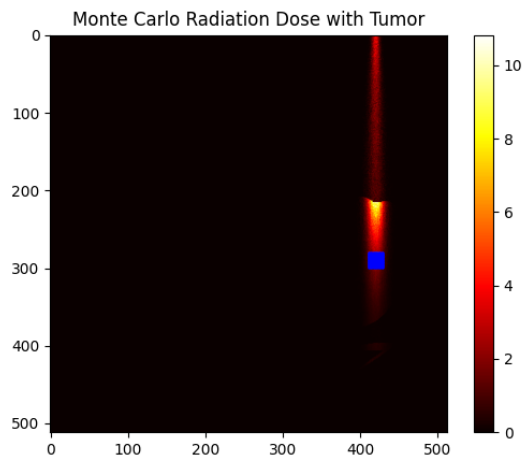

Fig. 10. Monte Carlo dose distribution with tumor overlay

7) Beam Strategy Comparison: A comparison between random and targeted beam strategies was performed. The measured mean tumor doses were:

- Random beam: 0.088
- Targeted beam: 3.59

Although no direct experimental validation is performed, the observed dose patterns are consistent with expected radiological behavior, including exponential attenuation and increased scattering in heterogeneous media.

VIII. Project Contributions

This project demonstrates the following contributions:

- Implementation of deterministic and stochastic photon transport models
- Integration of tissue segmentation using Hounsfield Units
- Quantitative comparison between random and targeted beam strategies
- Development of an interactive simulation interface enabling real-time exploration of radiation dose behavior

IX. Discussion

This study demonstrates the application of both deterministic and Monte Carlo approaches for modeling radiation dose deposition using CT-derived tissue information. The results highlight key differences in physical realism, computational behavior, and clinical relevance between the two models.

A. Deterministic vs Monte Carlo Modeling

The deterministic beam model, based on exponential attenuation, provides a simplified approximation of radiation transport. While computationally efficient, it assumes a straight-line trajectory with minimal scattering. As a result, it fails to capture the stochastic nature of photon interactions and produces overly smooth and unrealistic dose distributions.

In this work, dose values are expressed in arbitrary units proportional to relative energy deposition and are used for comparative analysis rather than absolute clinical calibration. The observed variability in dose reflects real-world radiation transport phenomena, where interactions depend on tissue composition and random scattering events.

B. Impact of Tissue Heterogeneity

The use of Hounsfield Units (HU) for tissue segmentation enabled differentiation between air, soft tissue, and bone. Each tissue type exhibited distinct attenuation behavior:

- Air showed low attenuation, allowing photons to penetrate deeper but deposit less energy locally.
- Soft tissue exhibited moderate attenuation and higher energy deposition, making it the primary region of dose absorption.
- Bone demonstrated high attenuation, significantly reducing photon penetration and resulting in minimal dose beyond dense regions.

These findings are consistent with known radiological properties and validate the use of CT-based attenuation mapping in dose simulations.

C. Tumor Dose Characteristics

The tumor dose analysis revealed significant spatial variability in dose deposition. Despite targeted beam delivery, the dose distribution within the tumor was not uniform. The computed underdose probability (approximately 0.40) indicates that a substantial portion of the tumor region received insufficient radiation.

This heterogeneity arises from stochastic photon interactions and limited beam spread. While targeting improves average tumor dose, it does not guarantee uniform coverage, which is critical for effective tumor control.

D. Effect of Beam Targeting

Targeted beam delivery significantly increased the mean tumor dose compared to random initialization. This demonstrates the importance of beam focusing in improving treatment effectiveness.

However, even with targeting, dose distribution remains uneven due to scattering and attenuation. In clinical practice, this limitation is addressed through techniques such as:

- Multi-angle beam delivery
- Intensity-modulated radiation therapy (IMRT)
- Volumetric modulated arc therapy (VMAT)

These approaches aim to achieve more uniform dose coverage while minimizing exposure to surrounding healthy tissue.

E. Limitations

Despite its strengths, this study has several limitations:

- The simulation is based on a single 2D CT slice rather than a full 3D volume.

- Tissue segmentation is simplified into only three classes, which does not capture finer anatomical detail.
- Photon interactions are modeled using simplified probabilistic rules rather than full physical cross-section data.
- Secondary particle effects (e.g., electron transport) are not included.
- Beam geometry is limited to a single-direction source without angular diversity.

Although these simplifications reduce computational complexity but limit the clinical accuracy of the model, they were intentionally introduced to balance computational tractability with physical interpretability.

F. Future Work

Future improvements to this work could include:

- Extension to 3D volumetric CT data for realistic patient modeling
- Incorporation of more detailed tissue classification and material properties
- Implementation of advanced Monte Carlo physics models with energy-dependent cross-sections
- Simulation of multiple beam angles and treatment planning strategies
- Integration of dose-volume histograms (DVH) for clinical evaluation

G. Clinical Relevance

Although simplified, this simulation captures the fundamental principles of radiation therapy planning. It demonstrates how tissue heterogeneity, beam targeting, and stochastic interactions influence dose distribution.

The results emphasize the importance of accurate modeling in ensuring adequate tumor coverage while minimizing damage to healthy tissue. Monte Carlo methods, despite their computational cost, remain the gold standard for high-fidelity dose calculation in modern radiotherapy systems.

X. Practical Applications

Although implemented in a simplified form, this simulation framework reflects core principles used in clinical radiotherapy planning systems.

Potential applications include:

- Educational tool for understanding radiation physics
- Prototype for treatment planning system development
- Benchmarking simplified dose calculation models
- Research platform for Monte Carlo algorithm experimentation

The framework can be extended toward clinically relevant systems such as IMRT and VMAT optimization engines.

Reproducibility

All simulations were implemented using standard scientific Python libraries. The methodology is fully reproducible given CT image input and parameter settings described in this work.

The workflow consists of CT data loading, HU conversion, tissue segmentation, deterministic or Monte Carlo simulation, and visualization. All parameters, including photon count, beam spread, and attenuation coefficients, are configurable.

XI. Conclusion

This study presented a CT-based radiation dose simulation framework integrating deterministic and Monte Carlo models. The results show that Monte Carlo simulation produces more physically realistic dose distributions by capturing stochastic photon interactions, while deterministic models provide a computationally efficient approximation based on exponential attenuation.

Targeted beam delivery significantly improves tumor dose compared to random beam placement; however, spatial dose heterogeneity remains due to scattering and tissue-dependent attenuation. This highlights the importance of stochastic modeling and beam optimization in radiation therapy planning.

Future work will extend this framework to full 3D volumetric CT data, incorporate more detailed physical interaction models, and explore integration with clinical treatment planning systems.

Overall, the framework demonstrates how medical imaging and Monte Carlo simulation can be combined into a unified computational approach for studying radiation dose deposition in heterogeneous media.

Disclaimer

This work is for educational and research purposes only and does not represent clinical treatment planning.

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